Machine Learning-Enabled Smart Transit: Real-Time Bus Tracking System for Enhanced Urban Mobility

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***Abstract*—In bustling urban cities, the efficiency of public transportation stands as a critical factor in alleviating traffic congestion and enhancing the daily commuter experience. This paper introduces a state-of-the-art Bus Tracking System that is meticulously designed to empower passengers by providing real-time updates on bus locations, availability, and estimated arrival and departure times. By leveraging cutting-edge technologies, this innovative system offers a myriad of advantages, notably including the reduction of passenger wait times, the optimization of route planning, and an overall enhancement of the public transportation journey. Through a comprehensive exploration of the significance of bus tracking technology and its potential to revolutionize urban mobility, this research underscores its pivotal role in promoting sustainable, convenient, and seamless transportation solutions tailored to the needs of modern cities. Our model uses machine learning techniques like regression, Kalman filter algorithm, and support vector machines to analyze the accuracy of arrival and departure of buses and further help in time-efficient and demand-based management of buses. A real-time location of buses can be provided by Google Maps API and Google Geolocation API.**

Keywords— Tracking, Machine Learning, Geo-Location, Kalman Filter, Support Vector Regression.

I. INTRODUCTION

Public transportation plays a pivotal role in addressing the growing challenges of urban mobility, such as traffic congestion, environmental pollution, and energy consumption. Monitoring and tracking public transport systems have become crucial to ensure efficient and reliable services for passengers. Real-time tracking of buses not only empowers commuters with valuable information but also allows transportation authorities to optimize fleet management and service planning[1].

With the ever-increasing urban population, there is a pressing need to improve the usage of public transport in cities. Promoting and encouraging the use of public transport can help reduce the number of private vehicles on the roads, easing traffic congestion and mitigating environmental impacts. An efficient and reliable bus tracking system holds the potential to significantly enhance public transport utilization by providing passengers with accurate and up-to-date information on bus locations, estimated arrival and departure times, and the availability of buses on various routes[12].

Machine Learning (ML) has emerged as a revolutionary technology with transformative potential across various domains. ML algorithms enable systems to learn from data, make informed predictions, and adapt their behavior accordingly. In transportation, ML algorithms can be harnessed to process vast amounts of real-time data generated by the bus tracking system. These algorithms can then analyze patterns, predict passenger demand, optimize bus schedules, and even suggest alternate routes during unforeseen events like traffic jams or road closures[16][17].

This paper presents a Machine Learning-Enabled Smart Transit solution, designed to revolutionize urban mobility by incorporating real-time bus tracking for passengers. Our proposed system harnesses the power of ML algorithms to process real-time data streams from buses, passengers, and external sources to provide accurate predictions on bus locations, arrival and departure times, and route availability[4]. By empowering commuters with this timely information, we aim to encourage higher adoption of public transport, leading to reduced traffic congestion, improved air quality, and a more sustainable urban transportation ecosystem.

Throughout this research, we delve into the intricacies of our ML-based bus tracking system, exploring various ML algorithms used for data analysis, prediction, and optimization. Additionally, we discuss the potential benefits and challenges associated with implementing such a system and its significance in promoting greener, more efficient, and passenger-friendly urban transport networks. By presenting this study, we hope to contribute to the advancement of smart and sustainable urban mobility

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solutions, addressing the needs of both passengers and public transportation authorities alike.

II. Literature Review

In recent years, researchers have recognized the significance of real-time bus tracking systems[1] in improving urban mobility and enhancing public transportation services. Various studies have explored different technologies and methodologies to address this issue.

One notable research study conducted by Rahul Bandhan, Saloni Garg, Assoc. Prof Bipin Kumar Rai, and Asst. Prof. Gaurav Agarwal [3] proposed a system utilizing Radio Frequency Identification (RFID) technology equipped with GPS modules. This system involved RFID readers integrated into buses, which read position coordinates from RFID tags provided to passengers. The implementation of this RFID-based system facilitated efficient bus tracking and management, enabling commuters to access accurate real-time bus information.

Several other researchers have also investigated innovative approaches to bus tracking and management. For instance, [21][25] presented a web-based application for bus tracking and management, aiming to enhance overall operational efficiency and passenger experience. Meanwhile, [7] proposed a GPS and IoT-based intelligent bus tracking system to optimize routes and improve the predictability of bus arrivals.

Machine Learning (ML) algorithms have emerged as a game-changer in various fields, including transportation. In the context of bus tracking, ML algorithms can be harnessed to analyze and predict bus locations and estimated arrival times more accurately. For example, [14] explored the application of ML techniques to predict bus arrival times, ensuring passengers have access to reliable and real-time information.

Our proposed model builds upon these existing studies by integrating ML algorithms into a Smart Transit solution. By employing real-time data streams[2] from buses and passengers, our system aims to provide passengers with up-to-date information on bus locations, estimated arrival and departure times, and route availability[22]. This ML-enabled approach strives to revolutionize urban mobility[20], promoting a more efficient and sustainable public transportation ecosystem while empowering commuters with a seamless travel experience[15].

III. Research Methodology

The web app was designed using react.js and the backend was designed using node.js. The User Interface of the web app is very simple and determines to show the location of buses and displays the routes of buses in different areas in a more interactive manner. JavaScript is widely used to extensively use the Google Maps API.

The web-based application in the front end is divided into two parts mainly for the user and driver. The user module mainly focuses on the login of the user and finding the best route to a desired destination.

The driver's module has a separate login through which the driver's permission is taken to retrieve the location of the bus using the Google Maps API or by providing local host connection.Usually these data points in which bus moves will be collected stored in database and In case of network issues machine learning algorithms are used to accurately predict the arrival times of buses.

The web application in the back end mostly focuses on the retrieval of location coordinates from the bus driver or conductor mobile and location coordinates are decoded to the server using Geo-coding API/Local Host connection. The location of the bus is tracked using Geo-location API which accepts the HTTP service request from the access point of a cell tower or WIFI Nodes of the mobile.

The Maps JavaScript API is used to display the location of Buses and their travel path is determined using the roads API. The Transit Layer object in Maps JavaScript API allows to display the public transport routes. Paths in Maps Static API is used to draw routes between two or more points for locating the bus route.

The traffic layer in Maps JavaScript API is used to determine the traffic in a particular route and the estimated time of arrivals (ETAs) will be determined. The geometry library in Maps JavaScript API is used to calculate the distances from the longitudes and latitudes retrieved by Geocoding API[21].

A MongoDB database is established to store the data of location coordinates and travel routes of buses in the form of geospatial data. The location coordinates longitudes and latitudes are stored in the form of coordinate pairs. The Travel routes can be represented on an earth-like sphere as GeoJSON objects.

The GeoJSON objects are loaded into the web app using the loadGeoJson() method from the data layer in Maps JavaScript API. The Transactions in the database were maintained using the Core API to start, commit, abort and manually deal with errors in the transactions received from the server[24]. The driver or conductor location coordinates are encrypted using in-use encryption which is encrypted to ciphertext data.

Our Machine Learning-Enabled Smart Transit system for real-time bus tracking encompasses various components that work harmoniously to provide passengers with accurate and timely information. Data collection is achieved by retrieving GPS data from drivers' or conductors' smartphones, which is then processed and linked using APIs to our backend system for analysis[9][10][11].

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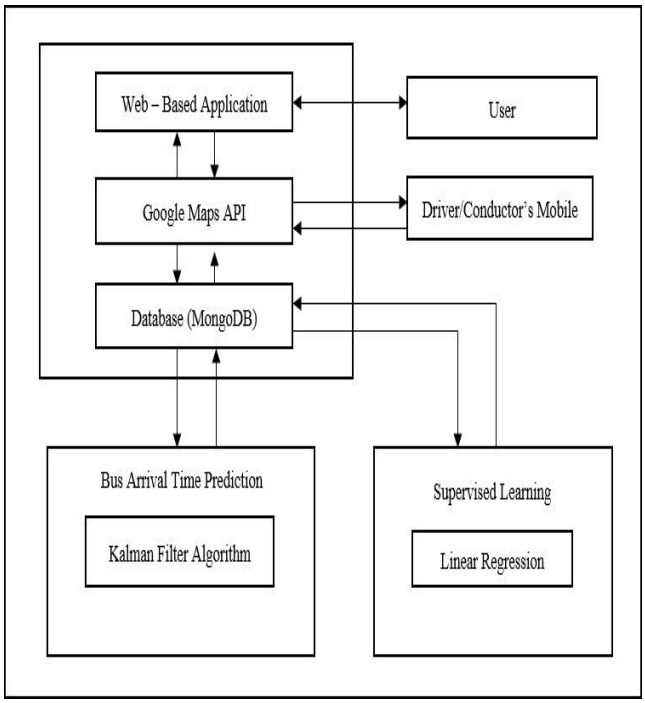


Fig. 1. Bus Tracking and Management System

Passengers access the system through an intuitive web application, where they can select their desired route, boarding, and destination addresses. The application displays a comprehensive list of available buses and their timings, enabling informed travel decisions.

To ensure passengers are well-prepared for their bus stops, our system incorporates a real-time timer feature[5], which alerts them five minutes prior to arrival.

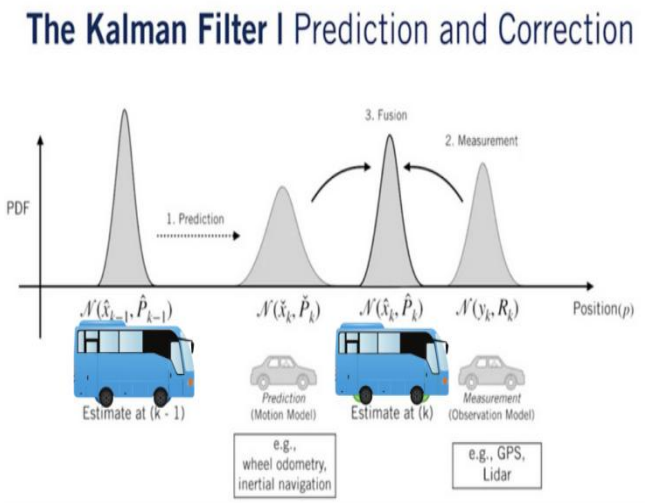


Fig. 2. Kalman Filter Application

Machine Learning algorithms are the cornerstone of our predictive capabilities. We draw inspiration from previous research [14], [7], and [3] and deploy the Kalman Filter, SVR, and kNN algorithms. The Kalman Filter estimates current bus locations and predicts future positions by refining location estimation and minimizing GPS signal irregularities.

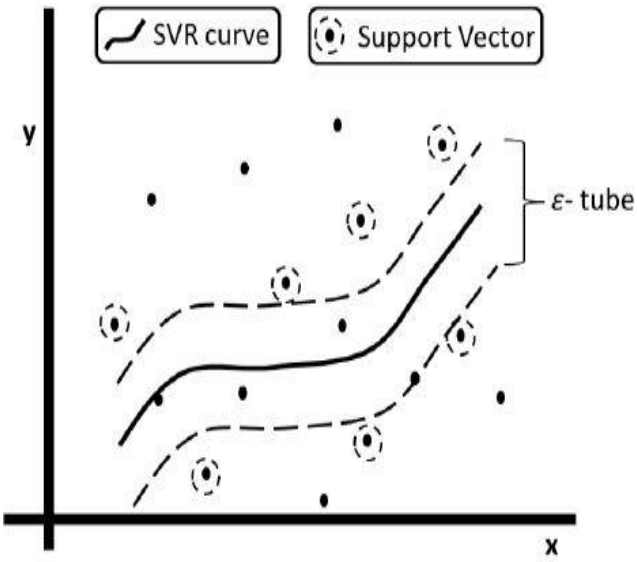


Fig. 3. Support Vector Regression

For regression analysis, we employ SVR to predict the time taken for buses to reach the next stop, utilizing historical latitude and longitude data points.

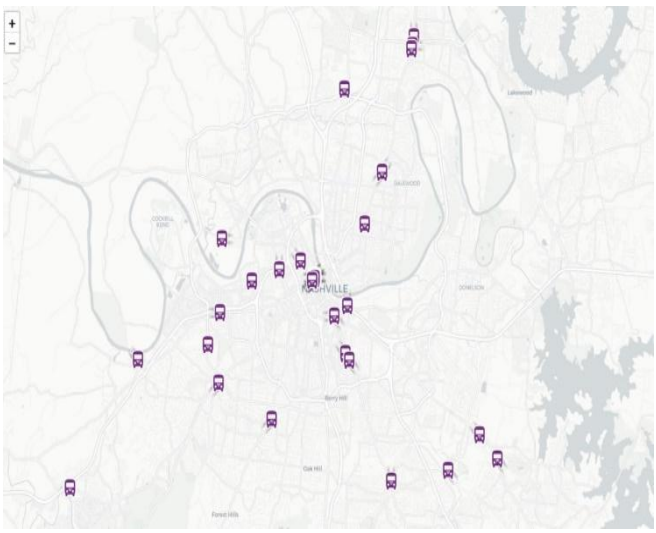


Fig. 4. Available buses and their position within the city

kNN aids in identifying similar data patterns, allowing for time estimation predictions based on neighboring bus trajectories.

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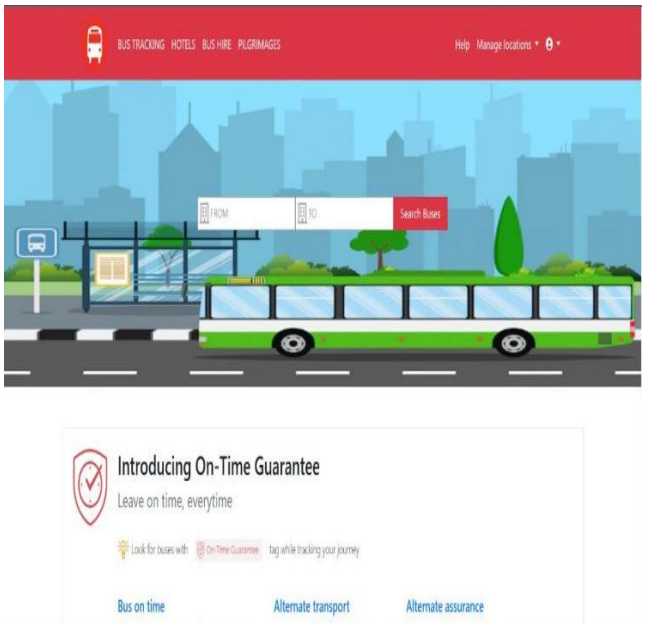


Fig. 5. Front-end Design for Website

The GPS data points collected from bus GPS readings are stored in our database, serving as inputs for the ML algorithm=ms. Continual learning is achieved by accumulating more data over time[6][8], enabling the ML model to improve accuracy.

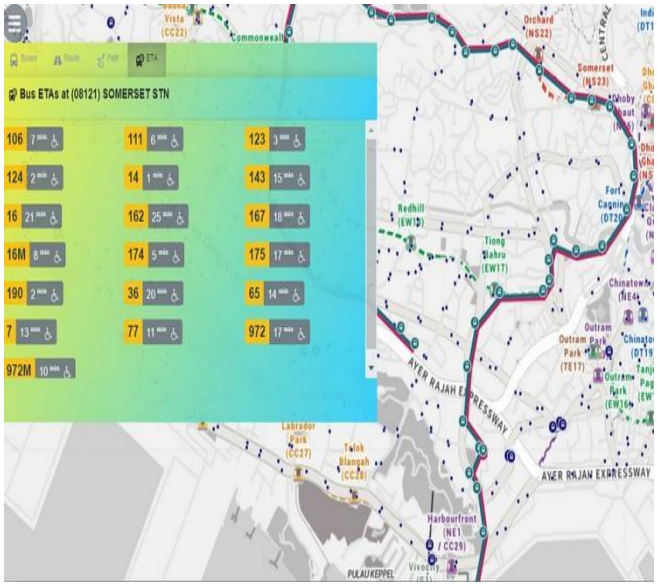


Fig. 6. Bus Arrival and Departure Times

To enhance the system's accuracy further, we adopt a reinforcement learning approach by soliciting post-journey feedback from passengers on predicted arrival times. This feedback loop refines outcomes and ensures an overall improved prediction accuracy [18][23] and [13].

By integrating these methodologies, our Machine Learning-Enabled Smart Transit system revolutionizes urban mobility, fostering the use of public transportation in cities while delivering precise and timely bus tracking information to passengers[19].

IV. Results

The developed model has been effectively validated and deployed, enabling users to track bus locations based on their current positions effortlessly. This model stands as a testament to creative ingenuity, meticulously crafted with minimal expenditure. Its impact is widespread, benefiting both the public and private transportation sectors. The model leverages cutting-edge technology, specifically Machine Learning, for real-time updates on bus locations and arrival projections.

With sufficient training, the model can operate autonomously without network connectivity for continuous updates and predictions. Notably, a distinctive feature is the incorporation of a timer mechanism that notifies passengers with an audible alarm before reaching their chosen destination.

This innovative approach showcases the potential to significantly enhance urban mobility and passenger experience, highlighting the prowess of modern technological advancements in transit systems.

V. Discussions

The implementation of our Machine Learning-Enabled Smart Transit system offers several significant advantages for urban mobility and public transportation services. Real-time bus tracking through GPS data ensures accurate and up-to-date information on bus locations, empowering passengers to plan their journeys efficiently.

The user-friendly interface provides easy access to bus schedules and improves passenger experience. ML algorithms, including Kalman Filter, SVR, and kNN, accurately predict bus arrival times, continually improving with data accumulation.

The reinforcement learning approach, driven by passenger feedback, refines the system's predictions. Overall, our system enhances public transport utilization and contributes to a more efficient and accessible urban transit network.

VI. Conclusion

In conclusion, our Machine Learning-Enabled Smart Transit system revolutionizes urban mobility by providing real-time bus tracking and accurate arrival predictions. Utilizing GPS data and advanced ML algorithms, our user-friendly interface empowers passengers to make informed travel decisions and enhances overall public transport experiences.

Continuous learning and passenger feedback further refine the system's predictive capabilities. With the potential to reduce congestion, promote sustainability, and improve accessibility, our research paves the way for a smarter and greener future in urban transportation.

References

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[1] R. Bandhan, S. Garg, B. K. Rai, G. Agarwal, "Real-Time Web-Based Bus Tracking System," International Research Journal of Engineering and Technology (IRJET), vol. 3, no. 4, pp. 20-24, Apr. 2016.

[2] A. Ghose and M. Sharma, "A Survey of Real Time Bus Arrival Time Prediction Models," arXiv preprint arXiv:1407.0313, 2014.

[3] S. Guduru and R. Sreeram, "GPS and RFID Based Real-Time Bus Tracking and Passenger Information System," 2017 International Conference on Trends in Electronics and Informatics (ICOEI), Chennai, India, 2017, pp. 581-584.

[4] T. Han and J. Yang, "GPS-Based Bus Arrival Time Prediction Using Gradient Boosting Decision Tree," Proceedings of the International Conference on Data Science and Advanced Analytics, 2017.

[5] M. Patel, P. Kumar, D. Thakkar, R. Shah, and H. Thakkar, "Real-Time Bus Tracking System," International Journal of Engineering Research & Technology (IJERT), vol. 9, no. 06, pp. 721-725, Jun. 2020.

[6] A. T. Mustafa, N. Khalid, A. Z. Azwandi, and N. A. Bakar, "Development of Smart Bus Tracking and Management System," Journal of Advanced Research in Dynamical and Control Systems, vol. 12, 2020.

[7] S. Kumar, A. Sharma, and V. Kumar, "IoT Based Bus Tracking and Alert System Using Raspberry Pi," 2020 6th International Conference on Advanced Computing and Communication Systems (ICACCS), Coimbatore, India, 2020, pp. 785-789.

[8] M. Srinivas, M. K. Chaitanya, and K. Kiran Kumar, "Real-Time Bus Tracking and Passenger Information System," 2018 International Conference on Advances in Computing, Communications and Informatics (ICACCI), Bangalore, India, 2018, pp. 1486-1490.

[9] A. Roy and K. Bhattacharya, "Design and Development of a Real-Time GPS-GPRS Based Vehicle Tracking and Fleet Management System," Electrical Engineering and Intelligent Systems, vol. 2, 2019.

[10] S. N. Chouhan and R. P. Singh, "GPS-GPRS Based Real-Time Bus Tracking and Passenger Information System," International Journal of Latest Engineering and Management Research, vol. 2, no. 2, 2019.

[11] S. B. Singh and P. Sharma, "Real-Time Bus Tracking and Passenger Information System," International Journal of Computer Science & Information Technology, vol. 8, no. 1, 2016.

[12] T. S. Kamal and V. Geetha, "IoT Based Smart Bus Tracking System," International Journal of Information Systems and Engineering, vol. 5, no. 3, 2019.

[13] A. Dutta, B. Roy, A. Saha, and A. Choudhury, "Real-Time Bus Tracking System using GPS and GSM Module," 2019 5th International Conference on Advanced Computing & Communication Systems (ICACCS), Coimbatore, India, 2019, pp. 79-83.

[14] M. Khan, M. Islam, and S. Saha, "Bus Tracking System using GPS and GSM Modem," 2015 International Conference on Electrical Engineering and Information Communication Technology (ICEEICT), Dhaka, Bangladesh, 2015, pp. 1-5.

[15] N. Vijayakumar, "Real-Time Bus Tracking and Arrival Time Prediction System," 2017 International Conference on Smart Technologies for Smart Nation (SmartTechCon), Bangalore, India, 2017, pp. 771-775.

[16] A. K. Jha, P. K. Sharma, and N. Kumar, "Smart Bus Tracking System using IoT and Android Application," 2017 2nd International Conference on Telecommunication and Networks (TEL-NET), Gwalior, India, 2017, pp. 168-171.

[17] N. A. Norizam, N. Z. Abdullah, and A. Kadir, "Development of a Real-Time Bus Tracking and Monitoring System using GPS and GSM Technology," 2019 IEEE Regional Symposium on Micro and Nanoelectronics (RSM), Putrajaya, Malaysia, 2019, pp. 135-140.

[18] T. R. Goyal, A. Sharma, and M. Rastogi, "Real-Time Bus Tracking System using GPS and GSM Modem," 2016 International Conference on Emerging Trends in Electrical, Electronics & Sustainable Energy Systems (ICETEESES), Dehradun, India, 2016, pp. 1-6.

[19] A. Al-Ameen, M. R. S. Sulaiman, and A. B. Al-Haiqi, "A Real-Time Bus Tracking System for Smart Cities using MQTT and Apache Kafka," 2019 IEEE 16th International Conference on Mobile Ad-Hoc and Sensor Systems (MASS), Monterey, CA, USA, 2019, pp. 282-290.

[20] S. Chettri and N. Ahamed, "Bus Tracking System using GPS and GSM Technology," 2018 International Conference on Information and Communication Technology (ICICT), Rourkela, India, 2018, pp. 1-4.

[21] R. Gang, D. Liu, and Y. Gao, "Research on Real-Time Bus Arrival Time Prediction Based on Internet of Things," 2018 IEEE 12th International Conference on Anti-counterfeiting, Security, and Identification (ASID), Xiamen, China, 2018, pp. 360-363.

[22] A. Khan, M. R. Islam, and M. Z. Islam, "A Smart Bus Tracking and Information System using GPS and GSM Technology," 2016 International Conference on Innovations in Science, Engineering and Technology (ICISET), Dhaka, Bangladesh, 2016, pp. 1-6.

[23] R. H. Yasin, A. M. Yusop, and K. Dimyati, "Mobile Bus Tracking System using GPS and GSM," 2018 5th International Conference on Computer Applications in Electrical Engineering - Recent Challenges of Electrical Engineering and their Impact on Technology Development, Pilsen, Czech Republic, 2018, pp. 1-6.

[24] S. Islam and M. S. Islam, "Real-Time Bus Tracking and Ticketing System using GPS and GSM," 2018 21st International Conference of Computer and Information Technology (ICCIT), Dhaka, Bangladesh, 2018, pp. 1-6.

[25] K. Singh and A. Sharma, "Real-Time Bus Tracking and Management using GPS and GSM Technology," 201

[26] M. I. Ahmed, G. Madhusudhan, N. M. K. Varma, T. Shalini, G. R. Babu, D. G. Arora, and F. Unnisa, "Web-Based Application for Bus Tracking and Management," International Research Journal of Modernization in Engineering Technology and Science, vol. 05, no. 05, pp. [PageRange], May 2023.

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